THE PERSISTENCE OF BANK PROFIT*

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January 2011

Abstract:

This paper examines the intensity of competition in 65 national banking industries. Country-level dynamic panel estimates of the persistence of bank profit are reported and compared. Persistence of bank profit is interpreted as an indicator of the intensity of competition, and as such is found to be consistent with traditional structure-based and conduct-based competition indicators. Persistence is negatively related to the rate of growth in GDP per capita, and positively related to the size of entry barriers. Persistence tends to be weaker, and competition stronger, in countries where institutional development is more advanced and external governance mechanisms are strong.

Banking, competition, dynamic panel estimation, entry, profitability, persistence
JEL: G21 L11

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*The authors are grateful to an anonymous referee for a number of helpful comments on a previous draft of this paper. The authors would also like to thank Roberto Violi, Francesco Cesarini, Giuseppe Zadra, Giorgio Gobbi, Barbara Casu, Franco Fiordilisi and Claudia Girardone. The financial support of the Istituto (formerly, Ente) Luigi Einaudi, sponsoring the research project "Competition in European Banking" is gratefully acknowledged. The usual disclaimer applies

The Persistence of Bank Profit

1. Introduction

The competitive landscape of banking has been transformed over the past three decades by deregulation, technological change and the globalization of goods and financial markets. These developments have impacted upon the operations, efficiency, productivity, margins and profitability of banks in all countries (Demirguc-Kunt and Huizanga, 1998; Claessens et al., 1998; Levine, 2003; Clarke et al., 2003; Carbo and Fernandez, 2007; Maudos and Fernanadez de Guevara, 2004; 2007; Berger, 2007).[1] Banks nowadays operate in product and geographical markets very different from those that existed thirty years ago, and have adopted a range of conventional and innovative technologies to serve their customers.

To date, most academic research on competition and its effects on bank performance is based on theoretical models that are static in nature (Claessens and Laeven, 2004; Shaffer 2004; Goddard and Wilson, 2009; Dick and Hannan, 2010). Static models are useful in identifying causal relationships between key variables when markets are in equilibrium, but they provide only a snapshot picture of a dynamic competitive process (Geroski, 1990). There is no certainty that conduct or performance measures observed at any point in time represent equilibrium values. For example, an empirical association between high concentration and high profitability, implicit in Structure Conduct Performance (SCP) models, might simply appear by chance, during a period when the relevant market is in a state of disequilibrium. If so, cross-sectional data does not capture (except by chance) the long-run equilibrium relationship. Furthermore, cross-sectional data usually does not contain sufficient information on which to base reliable policy decisions to promote competitive outcomes. An abnormal or monopoly profit realized in one period could disappear in the next, rendering intervention by government or regulatory agencies unnecessary.

This paper uses a simple dynamic model to test the hypothesis that competition eliminates any abnormal profit quickly, and bank profit rates converge rapidly towards their long-run equilibrium values. The alternative hypothesis is that the structural characteristics of banking markets in particular countries, or specialist knowledge or regulatory advantages enjoyed by incumbent banks, results in competition being impeded. The slower is the speed of adjustment, the stronger is the persistence of profit, and the greater is the extent of the departure from the competitive ideal.

This paper tests for the persistence of profit using bank-level data from 65 countries. Generalised Methods of Moments (GMM) estimation of a first-order autoregressive model for normalized profit rates yields estimates for the persistence of bank profit at country level. Factors that influence the degree of persistence are examined in a second-stage analysis. Persistence is negatively related to the rate of growth in GDP per capita, and positively related to the size of legal entry barriers. Persistence tends to be weaker in countries where institutional development is more advanced, and where mechanisms of external governance mechanisms are relatively strong.

The remainder of the paper is structured as follows. Section II reviews the theoretical and empirical literature on the persistence of profit. Section III describes the model specification. Section IV presents estimates of persistence for a sample of banks drawn from 65 countries. Section V examines the determinants of the persistence of profit. Section VI concludes.

2. Literature review

The persistence of profit approach is based on empirical investigation of the dynamics of firm-level profits. This constitutes a departure from the static, cross-sectional methodology that is prevalent in much of the literature based on the SCP paradigm and the New Industrial Organization (NEIO) approach.[2]

In its strongest form, the persistence of profit hypothesis developed by Mueller (1977, 1986) consists of two conditions. First, entry and exit are sufficiently free to ensure that competition eliminates any firm's abnormal profit quickly. Second, all firms' profit rates tend to converge towards the same long-run average value. The second condition is stronger than the first; and under a less restrictive version of the persistence of profit hypothesis, abnormal profit rates dissipate quickly and convergence is towards long-run average profit rates that may differ between firms. The empirical analysis in this paper adopts this latter formulation of the persistence of profit hypothesis. The alternative hypothesis is that some incumbent firms enjoy regulatory protection, or possess the capability to prevent imitation or block entry. If so, abnormal profit persists from year to year, and convergence is either slow or, in the most extreme case, non-existent.

A number of studies have tested the persistence of profit hypothesis for the (non-bank) manufacturing and services sectors (Mueller; 1986; Geroski and Jacquemin, 1988; Waring, 1996 Goddard and Wilson, 1996, 1999; McGahan and Porter, 1999; Glen et al., 2001, 2003; Marayuma and Odagiri 2002). This literature typically identifies patterns in the time-series variation of firm-level profit rate data, and draws inferences concerning the intensity of competition. Most studies report an average persistence of profit estimate, in the form of a first-order autoregressive coefficient in a time-series regression of firm-level normalized profit rates, in the range 0.4 to 0.5. A few studies have reported either lower or higher values. The persistence of profit has been found to be lower in developing countries than in developed countries. This pattern is attributed to lower sunk costs of entry, faster economic growth, and the presence of large conglomerate firms in developing countries, all of which may tend to increase the intensity of competition experienced by domestic firms (Glen et al., 2001; Stephan and Tsapin, 2008).

The evidence for the persistence of profit in banking is relatively scant.[3] Using a sample of large US banks observed over the period 1986-91, Levonian (1993) finds that despite restraints on competition imposed by bank regulation, abnormal profits tend to be temporary, rather than permanent. The speed of convergence, however, is slower than that suggested by most manufacturing studies. Roland (1997) examines the persistence of profit for US bank holding companies, using quarterly data for the period 1986-92. Persistence is stronger for banks with below-average performance. Berger et al. (2000) use non-parametric methods to examine the persistence of bank profit. The strength of persistence is found to differ between banks initially located in the top and bottom deciles of the distribution of banks by performance.

Goddard et al. (2004a) estimate persistence of profit coefficients for a sample of European banks from six countries, using a model that incorporates bank-specific variables including size, diversification, risk and ownership type. The persistence of profit is higher for mutual (savings and cooperative) banks than for commercial banks. By country, persistence is highest for France, where a strong regulatory tradition may have insulated banks from the full rigours of competition.

Several recent studies focus on individual countries. Agosttino et al. (2005) report estimates of persistence of profit coefficients for Italian banks for the period 1997-2000. Persistence is positively associated with ownership concentration. Knapp et al. (2006) report persistence estimates for a sample of US banks, suggesting that profits take about five years to converge towards average industry norms. Persistence estimates for Turkey and Greece are reported by Bektas (2007) and Athanasoglou et al. (2008), respectively. In a recent cross-country study for Sub-Saharan Africa, Flamini (2009) finds that strong persistence is associated positively with bank size, diversification and private ownership.

Overall, the empirical evidence to date focuses on a relatively small number of countries, and identifies positive autocorrelation in bank profit rates observed over time. The persistence of bank profit is driven by bank-specific and industry characteristics, and macroeconomic conditions. Surprisingly, there is little evidence on the extent to which regulation and supervision determine differences in the persistence of bank profit between countries. The principal objective of this paper is to fill this gap in the literature.

3. Methods

This section describes the empirical methods used in this paper. A two-stage approach is used to measure the persistence of profit, and investigate the factors that determine the persistence of profit. At the first stage, dynamic panel first-order autoregressive models are estimated for the normalized profit rates of banks in each of the 65 countries. At the second stage, the first-order autoregressive coefficients that were obtained at the first stage are used as the dependent variable in a cross-sectional regression for the determinants of the persistence of profit.

Stage 1 Modelling and estimating the persistence of profit

The normalized profit rate of bank i in year t is denoted $\pi_{i,t}$. The profit rate measure is return on average equity (ROE), measured by the ratio of net income after tax to total equity.[4] Taxes are deducted, because if effective tax rates differ between countries, entry and exit should be driven by after-tax profits. In order to eliminate the effects of cyclical fluctuations that impact similarly on the profit rates of all banks in the same country, the normalizing transformation expresses $\pi_{i,t}$ as a deviation from the cross-sectional mean profit rate in year t.

In the theoretical model, the change in the normalized profit rate of bank i between year t–1 and year t, denoted $\Delta\pi_{i,t}$, is assumed to be a function of the component of current and past entry that impacts directly on bank i's profitability, denoted $E_{i,t-k}$ for k=0 ... ∞; and an idiosyncratic component, denoted $u_{i,t}$:

$$\Delta\pi_{i,t} = \theta_i + \sum_{k=0}^{\infty}\beta_{j,k}E_{i,t-k} + u_{i,t} \tag{1}$$

In (1), the coefficients $\beta_{j,k}$, which reflect the impact of $E_{i,t-k}$ on $\Delta\pi_{i,t}$, are assumed to be the same for all banks located in country j. Entry is assumed to be a function of past realizations of bank i's normalized profit rate; and an idiosyncratic component, denoted $e_{i,t}$:

$$E_{i,t} = \phi + \sum_{k=1}^{\infty}\alpha_{j,k}\pi_{i,t-k} + e_{i,t} \tag{2}$$

In (2), the coefficients $\alpha_{j,k}$, which reflect the impact of $\pi_{i,t-k}$ on $E_{i,t}$, are assumed to be the same for all banks located in country j. Substituting (2) into (1) and re-parameterizing yields an autoregressive model for bank i's normalized profit rate:

$$\pi_{i,t} = \tilde{\pi}_i + \sum_{k=1}^{\infty}\lambda_{j,k}\pi_{i,t-k} + v_{i,t} \tag{3}$$

where $v_{i,t}$ is idiosyncratic. When employing panel data with a short time-dimension, it is convenient to adopt a first-order autoregressive (AR(1)) specification for $\pi_{i,t}$, with the higher-order lagged profit rates suppressed:

$$\pi_{i,t} = \tilde{\pi}_i + \lambda_j \pi_{i,t-1} + v_{i,t} \qquad (4)$$

In (4), $\tilde{\pi}_i$ denotes bank i's long-run mean normalized profit rate, and λ_j in (4) replaces $\lambda_{j,1}$ in (3). The adjustment of normalized bank profit rates described by (4) is interpreted as a consequence of the interaction between profitability and the entry threat, as postulated in the contestable markets literature (Baumol et al., 1982; Bratland, 2004).

Estimation of the persistence of profit coefficients, λ_j in (4), is implemented using Arellano and Bover's (1995) system GMM estimator, including both lagged differences and lagged levels of the dependent variable as instruments. The system GMM estimator reduces potential biases in finite samples, as well as the asymptotic imprecision that is associated with Arellano and Bond's (1991) difference GMM estimator (Blundell and Bond, 1998). The consistency of the system GMM estimator depends on both the validity of the assumption that the error term is free of second-order autocorrelation, and the validity of the instruments. Two specification tests are reported: a Hansen (1982) test for instrument validity, which is robust to heteroscedasticity in the disturbance term (Roodman, 2006, 2008); and a test of the null hypothesis of no second-order autocorrelation in the disturbance term.

Stage 2 Investigating the determinants of the persistence of profit

At stage 2, the determinants of variations in the country-level persistence of profit coefficients are examined. The estimated λ_j obtained from stage 1 forms the dependent variable in the stage 2 regression model, and country and banking sector characteristics are the covariates. The latter are grouped into the following four categories: macroeconomic and financial system; regulatory; institutional and external governance; and market structure and competition. Covariate definitions and sources are detailed in full in the Appendix.

The macroeconomic and financial system covariates are *inflation*, *market capitalization* and *GDP growth*. The banking market is less likely to be competitive when it is subject to high inflation, because the prices of financial services (retail banking interest rates, for example) are less informative (Claessens and Laeven, 2004) and may tend to exacerbate credit market frictions (Boyd et al., 2001). Angelini and Cetorelli (2003) find evidence of a negative relationship between inflation and competition, suggesting a positive association between high inflation and the persistence of profit.

Market capitalization, the aggregate value of all shares listed on the domestic stock market divided by GDP, is an indicator of the size of the stock market relative to the economy. Claessens and Laeven (2004) argue for a positive association between market capitalization and competition, on the basis that a highly developed non-banking financial sector tends to increase the competitive pressure on the banking sector. Accordingly, a negative relationship is expected between market capitalization and the persistence of profit.

A positive relationship is expected between *GDP growth* and business opportunities for banks, which might help banks to sustain positions of abnormal profitability. If so, a positive association would be expected between GDP growth and the persistence of profit. On the contrary, the availability of plentiful business opportunities might tend to strengthen competition between banks, in which case a negative relationship would be expected between GDP growth and the persistence of profit.

The regulatory control variables include proxies for legal barriers to entry, restrictions on bank activity and capital requirements, obtained from the Barth et al. (2004, 2006, 2008) database, and an index of financial freedom. The legal entry barriers measure *total entry denied* is the ratio of entry applications denied to total applications received from domestic and foreign banks during the period 2001 to 2005. A positive relationship with the persistence of profit is expected. *Domestic entry denied* and *foreign entry denied* are used as alternative legal entry barriers measures in separate estimations.

Activity restrictions is a measure of a bank's ability to provide fee-paying services related to securities, insurance and real estate, in addition to interest-based banking services.[5] A higher value indicates more onerous restrictions. Restrictions on permissible banking services offered might improve the safety and soundness of the banking system, by minimizing opportunities for banks to accept excessive risk, eliminating some conflicts of interest, and simplifying supervision. Such restrictions, however, might also reduce or eliminate opportunities for banks to realize economies of scale or scope, or manage the variability in earnings across product lines. *Capital requirements* is the minimum regulatory capital-to-assets ratio per country, and is also interpreted as an entry barriers measure. A positive association with the persistence of profit is expected. *Financial freedom* is the average value for the period 1997-2007 of an index of the extent of government regulation of financial services, state intervention in financial institutions, entry barriers to financial services, and government influence on the allocation of credit. A higher value signifies less freedom.

An association is expected between the quality of the country's institutions and external governance framework, and the persistence of bank profit. Governance rules and practice differ between countries, and are dependent on the level of economic and financial development (La Porta et al., 1998, 2002; Chhaochharia and Laeven, 2009). Four institutional and external governance covariates are considered, covering economic freedom, institutional development, GDP per capita, and property rights.

Economic freedom is the average value for the period 1997-2007 of an index of economic freedom (freedom from government interference afforded to businesses and individuals). A higher value signifies greater freedom. The *KKZ index* is the average value for the period 1998-2007 of an index of the level of institutional development. A higher value indicates a more advanced level of development. *GDP per capita*, expressed in natural logarithms, is a standard measure of economic development. Finally, protection of property rights is an important pre-requisite for a well-functioning financial system. *Property rights* is 100 *minus* the average value for the period 1997-2007 of a property rights protection index. A higher value signifies stronger protection of property rights.

Two market structure and competition covariates are employed. *HHI*, the average Herfindahl-Hirschman Index for each country's banking industry for the period 1997-2007, is used as an industry concentration indicator. According to the 'collusion' hypothesis, a positive relationship between industry concentration and the persistence of profit is expected (Berger, 1995; Goddard et al., 2001). According to the 'quiet life' hypothesis, on the contrary, a negative relationship between industry concentration and persistence of profit might be expected if the managers of large banks settle for a quiet life instead of pursuing profits (Hicks, 1935; Berger and Hannan, 1998). The Panzar and Rosse (1987) *H-statistic* is used as a conduct-based indicator of competition. The intensity of competition is inversely related to the magnitude of the H-statistic: $H<0$ indicates competitive conduct consistent with the theoretical model of monopoly; $0<H<1$ is consistent with monopolistic competition; and $H=1$ is consistent with perfect competition[6] (Carbo et al. 2009; Goddard and Wilson, 2009).

4. Data and Results

Financial accounts data (unconsolidated) are obtained from *Bankscope* for 11,634 banks from 65 countries, covering the period 1997 to 2007. The sample selection criteria are as follows. Any country with 15 or more commercial banks (domestic or foreign) is selected. The start-year, 1997, is constrained by the availability of data on *Bankscope*. To eliminate outliers, observations for which the profit rate was more than three standard deviations from the sample mean for each country are deleted**.** The final sample for the first-stage analysis is an unbalanced panel with 46,959 bank-year observations (11,634 banks).

Columns (1) to (4) of Table 1 report descriptive statistics for ROE by country, classified by International Monetary Fund (2008) as advanced or developing. Average profitability varies significantly between countries. The lowest average ROE are -9.10% (Uruguay) and -4.29% (Thailand). The highest are 25.37% (Venezuela) and 23.84% (Nigeria). Average profitability was higher in developing countries (10.96%) than in advanced countries (7.99%).

First-stage analysis: estimating the persistence of profit

Table 1 reports estimates of the short-run persistence of profit coefficients, λ_j in (5), by country. The Hansen and second-order autocorrelation tests suggest that the estimations reported in Table 1 are, in general, appropriately specified. The relatively small numbers of rejections of the null hypothesis can reasonably be attributed to Type I error.

The estimated persistence of profit coefficients are positive for 62 of the 65 countries, and negative for three countries (Croatia, Dominican Republic and South Korea). For 40 of the 65 countries, the estimated λ_j lies within the interval 0.2 to 0.6. The mean λ_j for the entire sample is 0.43, somewhat higher than the mean value for six European countries of 0.26 reported by Goddard et al. (2004a). The results suggest that the elimination of abnormal profits through competition is by no means instantaneous, and banks are able to retain a significant portion of their abnormal profits from year to year.

Table 2 reports the frequency distribution of the estimated persistence of profit coefficients, λ_j for advanced and developing countries. On average advanced countries have slightly higher persistence of profit (average λ_j = 0.442) than developing countries (average λ_j = 0.426), but the difference between the mean estimated λ_j is not statistically significant.

Second-stage analysis: identifying the determinants of persistence of profit

The second-stage analysis examines the determinants of the persistence of profit. Table 3 reports descriptive statistics for the entire sample for ROE, the estimated persistence coefficients (the dependent variable in the second-stage regressions), and all covariates used in the second-stage regressions. Table 4 reports the correlation matrix for the covariates.

Tables 5 and 6 report the estimation results. In model (1), only the macroeconomic covariates (inflation, GDP growth, stock market capitalization) are included in the estimation. These covariates are also included as controls in all of the other estimations. In models (2) to (7), the regulatory covariates (total entry denied, domestic entry denied, foreign entry denied, activity restrictions, capital requirements, financial freedom) are included, each in turn. In models (8) to (11), the institutional and external governance covariates (economic freedom, KKZ index, property rights, GDP per capita) are included in turn. In models (12) and (13), finally, the market structure and competition covariates (HHI, H-statistic) are included in turn.

Among the macroeconomic variables, the coefficients on the inflation covariate are generally insignificant. In column (6), there is a significant negative relationship between stock market capitalization and the persistence of profit. This would suggest that the more highly developed is the non-bank financial sector, the greater is the competitive pressure on the banking sector. Overall, however, there is little evidence for any relationship between financial sector development and the persistence of bank profit. The coefficients on GDP growth are predominantly negative, and significant in many cases. This is consistent with the hypothesis that business opportunities afforded by higher economic growth lower the opportunity cost of entering the banking sector, thereby reducing entry barriers. Stronger competition weakens the ability of incumbent banks to sustain their profitability.

The coefficients on the three legal barriers to entry covariates in columns (2), (3) and (4) are positive and significant. This suggests that successful banks in markets that are protected by entry barriers are better able to sustain their profitability over time. The remaining regulatory variables - activity restrictions, capital regulation, and financial freedom - appear to exert little or no influence on the persistence of bank profit. Overall, regulation appears to impact on the persistence of profit through the erection and enforcement of legal barriers to entry. For incumbent banks, however, other forms of regulation do not have any discernable effect on competition measured by the persistence of profit.

Among the institutional and external governance covariates, the coefficient on the economic freedom measure is negative and significant (column (8)), the coefficient on the KKZ index is negative and significant (column (9)), and the coefficient on the property rights measure is negative and significant (column (11)). *Ceteris paribus* the persistence of profit is higher in countries with less freedom from government interference, in countries at a lower level of institutional development, and in countries with weaker protection of property rights. The latter result is consistent with the hypothesis that when the protection of property rights is strong, new entrants feel more secure in conducting business and the intensity of competition increases.

Finally, among the market structure and competition covariates, the relationship between HHI and the persistence of profit is positive and significant (column (12)), and the relationship between the H-statistic and the persistence of profit is negative and significant (column (13)). The interpretation of the persistence of profit coefficient as a measure of the intensity of competition is therefore consistent with traditional structure-based and conduct-based competition indicators.

In summary, the empirical evidence suggests that the persistence of bank profit is negatively related to the rate of growth in GDP per capita, suggesting that the business opportunities afforded by higher economic growth tend to enhance competition and weaken the ability of incumbent banks to sustain abnormal profits. The persistence of bank profit is positively related to the size of legal entry barriers, in accordance with the view that actual or potential entry is a key determinant of the intensity of competition. There is an association between several institutional and external governance covariates and the persistence of bank profit: the latter is higher where businesses and individuals are afforded less freedom from government interference, where the level of institutional development is low, and where the protection of property rights is relatively weak. Finally, and in accordance with the interpretation of the persistence coefficient as an indicator of the intensity of competition, the persistence of bank profit is positively related to the level of industry concentration, and negatively related to a conduct-based measure of competitive conditions.

5. Conclusion

Over the past three decades, the banking industry has undergone fundamental regulatory and structural change, with profound implications for the intensity of competition. Using a large cross-country dynamic panel data set, this paper reports estimates of the persistence of bank profit. In contrast with the essentially static Structure Conduct Performance (SCP) paradigm and New Empirical Industrial Organization (NEIO) approach, which dominate the analysis of competition in the empirical banking literature, this paper adopts a dynamic view of competition. The present approach is consistent with the view, articulated in the contestable markets literature, that threatened or potential entry is as important as actual entry in determining the intensity of competition.

A two-stage analysis of the persistence of profit is reported using data on banks from 65 countries over the period 1997 to 2007. At the first stage, a first-order autoregressive model is used to estimate the short-run persistence of bank profit. The speed of convergence varies between countries, with banks in advanced countries exhibiting slightly higher persistence on average than those in developing countries; the difference, however, is not statistically significant. Although competition exerts some discipline on bank profitability, the adjustment towards the competitive equilibrium is by no means instantaneous. Impediments to competition, such as entry barriers, are sufficient to enable banks to retain a significant portion of their abnormal profits from year to year.

At the second stage, the determinants of the persistence of profit are examined. The estimated persistence coefficients from the first stage are used as the dependent variable in regressions with covariates that reflect macroeconomic conditions, the regulatory environment, institutional development and external governance, and market structure and competition. The persistence of bank profit is negatively related to the rate of growth in GDP per capita, and positively related to the size of entry barriers. There is an association between several institutional and external governance covariates and the persistence of bank profit: persistence tends to be weaker, and competition more intense, in countries where institutional development is further advanced and external governance mechanisms are stronger. Finally, the persistence of bank profit is positively related to the level of industry concentration, and negatively related to a conduct-based measure of competitive conditions.

From an antitrust, regulatory and supervisory perspective, a dynamic model of profitability provides an indication of the effectiveness of competition in forcing the adjustment or convergence of profits (above or below the norm) towards their long-run equilibrium values. This assists the regulator in distinguishing between cases in which a competitive equilibrium is likely to be achieved rapidly without intervention, and cases in which regulatory intervention may be required in order to achieve a competitive outcome.

Appendix Variable definitions and sources

Return on average equity	Total net income after tax, divided by average equity.[1]
Inflation	Average annual first difference of log GDP deflator over the period 1997 to 2007.[2]
Market capitalization	Average ratio of stock market capitalization to GDP, 1997-2007.[2]
GDP growth	Average annual first difference of log real GDP, 1997-2007.[2]
Total entry denied	Ratio of entry applications denied to applications received from domestic and foreign banks during the period 2001-2005.[3]
Domestic entry denied	As above, for domestic applications only.[3]
Foreign entry denied	As above, for foreign applications only.[3]
Activity restrictions	Categorical scale from 0 to 12 at year-end 2005: a higher value indicates more onerous restrictions.[4]
Capital requirement	Minimum regulatory capital-to-assets ratio per country at year-end 2005.[5]
Financial freedom	Average value for the period 1997-2007 of 100 *minus* the financial freedom index of the Heritage Foundation. A higher value signifies less freedom or more regulation.[6]
Economic freedom	Average value for the period 1997-2007 of the Heritage Foundation index of the degree of freedom from government interference afforded to businesses and individuals. A higher value indicates greater freedom.[6]
KKZ Index	Average value for the period 1998-2007 of an index of the level of institutional development, based on six dimensions of

	governance. A higher value indicates a more advanced level of development.[7]
GDP per capita	average log ratio of GDP to population, 1997-2007.[2]
Property rights	100 *minus* the average value for the period 1997-2007 of the Heritage Foundation property rights protection index. A higher value signifies greater protection. [6]
HHI	average Herfinhahl-Hirschman Index, defined as the sum of the squared market shares of all banks per country, 1997-2007.[1]
H-statistic	Sum of estimated coefficients on factor input prices in linear regression for log revenue[1] (see footnote 6).

Sources

1. *Bankscope*.
2. World Bank Economic Development Indicators.
3. Questions 1.9.1 to 1.9, 1.10.1 to 1.10, World Bank Survey of Bank Regulation and Supervision (for details see Barth et al., 2008).
4. Questions 4.1 to 4.4, World Bank Survey of Bank Regulation and Supervision.
5. Question 3.1, World Bank Survey of Bank Regulation and Supervision (for details see Barth et al., 2008).
6. Heritage Foundation (2009).
7. Kaufman et al. (2008).

References

Agostino, M., Leonida, L., Trivieri, F., 2005. Profits persistence and ownership: evidence from the Italian banking sector. Applied Economics 37, 1615-21.

Angelini, P., Cetorelli, N., 2003. Effects of regulatory reform on competition in banking, Journal of Money, Credit and Banking 35, 663-684.

Arellano, M., Bond, S., 1991. Some tests of specification for panel data: Monte carlo evidence and an application to employment equations. Review of Economic Studies 58, 277–297.

Arellano, M., Bover, O., 1995. Another look at the instrumental variable estimation of error-components models. Journal of Econometrics 68, 29-51.

Athanasoglou, P.P., Brissimis, S.N., Delis, M.D., 2008. Bank-specific, industry-specific and macroeconomic determinants of bank profitability. Journal of International Financial Markets Institutions and Money 18, 121-136.

Barth, J.R., Caprio, G., Levine, R., 2004. Bank regulation and supervision: what works best? Journal of Financial Intermediation 13, 205-248.

Barth, J.R., Caprio, G., Levine, R., 2006. Re-thinking Bank Regulation: Till Angels Govern. Cambridge: Cambridge University Press.

Barth, J.R., Caprio, G., Levine, R., 2008. Bank regulations are changing: For better or worse? Comparative Economic Studies 50, 537-63

Baumol, W., Panzar, J., Willig, R., 1982. Contestable Markets and the Theory of Industrial Structure. New York: Harcourt Brace Jovanovich.

Beck, T., Demirguc-Kunt, A., Maksimovic, V. 2004. Bank competition and access to finance. Journal of Money Credit and Banking 36, 627-648.

Bektas, E., 2007. The persistence of profits in the Turkish banking system, Applied Economics Letters 14, 187-190.

Berger, A.N., 1995. The profit-structure relationship in banking: tests of market power and efficient structure hypotheses. Journal of Money, Credit and Banking 27, 404-431.

Berger, A.N., 2007. Obstacles to a global banking system: Old Europe versus new Europe, Journal of Banking & Finance 31, 1955-1976.

Berger, A.N., Hannan, T.H. 1998. The efficiency cost of market power in the banking industry: a test of the quiet life and related hypotheses. Review of Economics and Statistics 80, 454–65.

Berger, A.N., Kashyap, A.K., Scalise, J.M., 1995. The transformation of the US Banking industry: what a long strange trip it's been. Brookings Papers on Economic Activity 2, 54-219.

Berger, A.N., Bonime, S.D., Covitz, D.M., Hancock, D., 2000. Why are bank profits so persistent? The roles of product market competition, information opacity and regional macroeconomic shocks. Journal of Banking & Finance 24, 1203-1235.

Berger, A.N., Demerguc-Kunt, A., Levine, R., Haubrich, J.C., 2004. Bank concentration and competition. Journal of Money, Credit and Banking 36, 433-451.

Black, S.E., Strahan, P.E., 2002. Entrepreneurship and bank credit availability, Journal of Finance, 57, 2807-2833.

Blundell, R., Bond S., 1998. Initial conditions and moment restrictions in dynamic panel data models. Journal of Econometrics 87, 115-143.

Bonaccorsi Di Patta, E., Dell'Aricca, G., 2004. Bank competition and firm creation, Journal of Money, Credit and Banking 36, 225-252.

Boyd, J.H., Levine, R., Smith, B.D., 2001. The impact of inflation on financial sector performance. Journal of Monetary Economics 47, 221-248.

Bratland, J., 2004. Contestable market theory as a regulatory framework: An Austrian post-mortem. Quarterly Journal of Austrian Economics 7, 3-28.

Carbo, S., Fernandez, F., 2007. The determinants of bank margins in European banking, Journal of Banking & Finance, 31 2043-2063.

Carbo, S. Humphrey, D. Maudos, J., Molyneux, P., 2009. Cross-country comparisons of competition and pricing power in European banking, Journal of International Money and Finance 28 115–134

Cetorelli, N., Strahan, P.E. 2006. Finance as a barrier to entry: bank competition and industry structure in local US banking markets, Journal of Finance, LXI, 437- 461.

Chhaochharia, V., Laeven, L., 2009. Corporate governance norms and practices, Journal of Financial Intermediation 18, 405-431.

Clarke, G., Cull, R., Martinez Peria, M.S., Sanchez, S.M., 2003. Foreign bank entry: Experience, implications for developing economies, and agenda for further research. World Bank Research Observer 18, 25-59.

Claessens, S., Demirguc-Kunt, A., Huizinga, H., 1998. How does foreign entry affect the domestic banking sector? World Bank Working Paper No: 1918.

Claessens, S., Laeven, L., 2004. What drives bank competition? Some international evidence. Journal of Money, Credit and Banking 36, 563-83.

DeYoung, R., Hunter, W., Udell, G., 2004. The past, present and probable future for community banks. Journal of Financial Services Research, 25, 85-133.

Demirguc-Kunt, A., Huizinga, H., 1998. Determinants of commercial bank interest margins and profitability: some international evidence. World Bank Economic Review 13, 379-408.

Dick, A.A., Hannan, T.H., 2010. Competition and antitrust in banking, in Berger, A.N., Molyneux, P. and Wilson, J.O.S. (eds.) Oxford Handbook of Banking. Oxford: Oxford University Press.

Flamini, V., McDonald, C.A., Schumacher, L.B., 2009. The determinants of commercial bank profitability in Sub-Saharan Africa, IMF Working Paper No. 09/15.

Geroski, P.A., 1990. Modelling persistent profitability, in Mueller, D.C. (ed.) The Dynamics of Company Profits: An International Comparison. Cambridge: Cambridge University Press.

Geroski, P.A., Jacquemin, A., 1988. The persistence of profits: a European comparison, Economic Journal 98, 375-89.

Glen, J., Lee, K., Singh, A., 2001. Persistence of profitability and competition in emerging markets: a time series analysis. Economic Letters 72, 247-53.

Glen, J., Lee, K., Singh, A., 2003. Corporate profitability and the dynamics of competition in emerging markets: a time series analysis, Economic Journal 113, F465-F484.

Goddard, J.A., Wilson, J.O.S., 1996. Persistence of profits for UK manufacturing and service sector firms. Service Industries Journal 16, 105–17.

Goddard, J. and Wilson, J.O.S. 1999. Persistence of profit: a new empirical interpretation. International Journal of Industrial Organization 17, 663-87.

Goddard, J., Molyneux, P., Wilson, J.O.S., 2001. European Banking: Efficiency, Technology and Growth. John Wiley, Chichester, UK.

Goddard, J., Molyneux, P., Wilson, J.O.S., 2004a. The profitability of European banks: A cross-sectional and dynamic panel analysis. Manchester School 72, 363-381.

Goddard, J., Molyneux, P., Wilson, J.O.S., 2004b. Dynamics of growth and profitability in banking. Journal of Money, Credit and Banking 36, 1069-1090.

Goddard, J., Molyneux, P., Wilson, J.O.S., Tavakoli, M., 2007. European Banking: an overview, Journal of Banking and Finance 31, 1911-1936.

Goddard, J., McKillop, D.G. Wilson, J.O.S., 2008. The diversification and financial performance of US Credit Unions. Journal of Banking and Finance 32, 1836-49

Goddard, J., Wilson, J.O.S., 2009. Competition in banking: a disequilibrium approach, Journal of Banking & Finance 33, 2282-2292.

Hansen, L., 1982. Large sample properties of generalized method of moments estimators, Econometrica 50, 1029–1054.

Heritage Foundation, 2009. Index of Economic Freedom, 2009, Methodology for the 10 Economic Freedoms.

Hicks, J., 1935. Annual survey of economic theory: The theory of monopoly. Econometrica 3, 1-20.

International Monetary Fund. 2008. International Financial Statistics. Washington: International Monetary Fund, Washington.

Jones, K.D., Critchfield, T., 2005. Consolidation in the US banking industry: is the ‘long, strange trip’ about to end? FDIC Banking Review 17, 31-61.

Kaufmann D., Kraay, A., Mastruzzi, M., 2008. Governance matters VII: Governance indicators for 1996-2007. World Bank Policy Research Working Paper, Number 4654.

Knapp, M., Gart, A., Chaudhry, M., 2006. The impact of persistence and mean reversion of bank profitability on post-merger performance in the banking industry. Journal of Banking & Finance 30, 3503-3517.

La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R., 1998. Law and finance. Journal of Political Economy 106, 1113–55.

La Porta, R., Lopez-de-Silanes, F., Shleifer, A., Vishny, R., 2002. Investor protection and corporate valuation. Journal of Finance 57, 1147–70.

Levine R., 2003. Denying foreign bank entry implications for bank interest margins. Central Bank of Chile Working Papers, Number 222.

Levonian, M.E., 1993. The persistence of bank profits: what the stock market implies. Working Papers in Applied Economic Theory, No. 93-15. San Francisco, Federal Reserve Bank of San Francisco.

Marayuma, N., Odagiri, H., 2002. Does the persistence of profits persist? A study of company profits in Japan, 1964-1997, International Journal of Industrial Organization 20, 1513-1533.

Maudos, J., Fernandez de Guevara, J., 2004. Factors affecting the interest margin in the banking sectors of the European Union, Journal of Banking & Finance 28, 2259-81.

Maudos, J., Fernandez de Guevara, J., 2007. The cost of market power in banking: social welfare versus cost inefficiency. Journal of Banking & Finance 31, 2103-2126.

McGahan, A.M., Porter, M.E., 1999. The persistence of shocks to profitability. Review of Economics and Statistics 81, 143-153.

Mueller, D.C., 1977. The persistence of profits above the norm. Economica 44, 369-380.

Mueller, D.C., 1986. Profits in The Long Run. Cambridge University Press, Cambridge, UK.

Panzar, J.C., Rosse, J.N., 1987. Testing for monopoly equilibrium. Journal of Industrial Economics 35, 443-456.

Roland, K.P., 1997. Profit persistence in large U.S. bank holding companies: an empirical investigation, Economics Working Paper, 97-2. Office of the Comptroller of Currency.

Roodman, D., 2006. How to do xtabond2: an introduction to “difference” and “system” GMM in Stata. Center for Global Development Working Paper Number 103.

Roodman, D., 2008. A note on the theme of too many instruments, Center for Global Development Working Paper Number 125.

Shaffer, S., 2004. Patterns of competition in banking, Journal of Economics and Business 56, 287-313.

Stephan, A., Tsapin, A., 2008. Persistence and determinants of firm profit in emerging markets. Applied Economics Quarterly 54, 231-253.

Stiroh, K., Strahan, P.E., 2003. Competitive dynamics of deregulation: evidence for US banking. Journal of Money, Credit and Banking 35, 801-28.

Stiroh, K.J., 2010. Diversification in banking, in Berger, A., Molyneux, P. and Wilson, J.O.S. (eds.) Oxford Handbook of Banking. Oxford: Oxford University Press.

Stiroh, K.J., Rumble, A., 2006. The dark side of diversification: The case of US financial holding companies. Journal of Banking & Finance 30, 2131-2432.

Vander Vennet, R., 2002. Cost and profit efficiency of financial conglomerates and universal banks in Europe. Journal of Money, Credit and Banking 34, 254-82.

Waring, G., 1996. Industry differences in the persistence of firm-specific returns, American Economic Review 86, 1253-65.

Zarutskie, R. 2006. Does bank competition affect how much firms can borrow? Journal of Financial Economics 81, 503-537.

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Footnotes

1. Berger et al. (1995) and Jones and Critchfield (2005) survey structural and strategic change in US banking, and Goddard et al. (2007) provide an overview of European banking. DeYoung et al. (2004) and Berger et al. (2004) analyze the responses of banks to changes in the competitive environment, and Vander Vennet (2002), Stiroh and Strahan (2003) and Maudos and Fernandez de Guevara (2004) examine the impact on performance. Beck et al. (2004) and Cetorelli and Strahan (2006) analyse the effect of bank competition on the availability of credit to small, medium and large firms. Black and Strahan (2002), Bonaccorsi Di Patta and Dell'Aricca (2004) and Zarutskie (2006) assess the implications for borrowing, investment, entrepreneurship and new firm creation.

2. For an overview, see Berger (1995), Berger et al. (2004), Dick and Hannan (2010), Goddard et al. (2001, 2004b), Claessens and Laeven (2004), Shaffer (2004).

3. Stiroh and Strahan (2003) test the effectiveness of deregulation in strengthening the competitive process. Deregulation increases both entry and exit, and reallocates market share from unprofitable banks to profitable banks.

4. The empirical analysis is also conducted using a return on average assets (ROA) profitability measure. To save space these estimations are not reported. They are available from the authors upon request.

5. In recent years banks in both developed and developing countries have increased the proportion of their operating income derived from non-interest sources. Diversification into lines of business such as investment banking, venture capital and insurance underwriting has contributed to this trend. Growth in fee-paying and commission-paying services linked to traditional retail banking services has also been important. Empirical evidence suggests that the shift towards non-interest income has not improved risk-adjusted returns (Stiroh and Rumble, 2006; Goddard et al., 2008; Stiroh, 2010).

6. The H-statistic is estimated using the following reduced-form revenue equation:

$$p_{i,t} = \alpha + \beta_1 w_{1,i,t} + \beta_2 w_{2,i,t} + \beta_3 w_{3,i,t} + \gamma_1 y_{1,i,t} + \gamma_2 y_{2,i,t} + \gamma_3 y_{3,i,t} + \sum_j \delta_j D_{j,t} + \varepsilon_{i,t}$$

where $p_{i,t}$ is the ratio of gross interest revenue to total assets (proxy for output price of loans) for bank i in year t. The factor input prices are: $w_{1,i,t}$ = ratio of interest expense to total deposits and money expense to total assets (proxy for input price of deposits); $w_{2,i,t}$ = ratio of personnel expenses to total assets (proxy for input price of labour); $w_{3,i,t}$ = ratio of overheads to total assets (proxy for input price of equipment/fixed capital). The control variables are: $y_{1,i,t}$ = ratio of equity to total assets; $y_{2,i,t}$ = ratio of net loans to total assets; $y_{3,i,t}$ = total assets. All variables are in logs. $D_{j,t}$ are year dummies for years j=1998,...,2007. The H-statistic is $H = \beta_1 + \beta_2 + \beta_3$.

Table 1 First-stage estimation results: ROE and short-run persistence by country

	Banks	Obs.	ROE mean	ROE s.d.	$\hat{\lambda}_j$	se($\hat{\lambda}_j$)	Hansen	AR(2)
Advanced countries								
Australia	26	87	11.52	21.86	.255	.085	.477	.507
Austria	248	1361	7.10	8.08	.477	.039	.570	.107
Belgium	62	344	9.80	19.07	.539	.004	.522	.547
Cyprus	15	71	16.17	28.65	.508	.011	.990	.136
Denmark	104	636	10.70	7.92	.488	.010	.105	.990
France	343	2022	9.17	19.08	.446	.024	.331	.031
Germany	2123	12178	5.41	5.57	.178	.017	.000	.002
Ireland	25	127	9.43	9.51	.779	.011	.850	.623
Italy	671	1863	7.13	7.91	.464	.029	.272	.623
Japan	758	4711	-1.94	19.06	.132	.031	.143	.471
Luxembourg	126	702	12.86	12.46	.425	.035	.439	.614
Netherlands	46	219	11.20	11.20	.719	.005	.785	.277
Norway	105	420	8.64	5.17	.375	.001	.461	.499
Portugal	24	106	9.91	9.95	.482	.009	.999	.361
Spain	92	428	8.73	13.76	.531	.015	.332	.553
Sweden	91	466	7.91	5.00	.118	.023	.065	.600
Switzerland	442	2392	8.18	7.51	.725	.032	.116	.861
United Kingdom	145	692	9.13	13.47	.684	.017	.455	.198
United States	738	4319	12.76	9.27	.680	.041	.098	.004
Developing countries								
Argentina	94	543	-3.29	37.14	.164	.008	.302	.546
Bahamas	16	89	19.83	17.67	.353	.098	.988	.294
Bangladesh	31	198	17.94	17.03	.284	.034	.192	.811
Bosnia-Herz.	23	124	6.28	10.09	.525	.033	.474	.313
Brazil	151	861	14.14	23.34	.311	.021	.102	.818
Bulgaria	22	132	13.15	16.33	.392	.008	.444	.220
Cayman Islands	16	82	11.29	11.48	.613	.046	.999	.387
Chile	34	197	9.26	10.76	.400	.005	.474	.451
China	70	345	12.78	10.08	.570	.025	.823	.727
Colombia	27	169	9.39	30.87	.324	.003	.957	.449
Costa Rica	45	208	13.14	7.14	.887	.016	.866	.832
Croatia	41	226	8.03	12.92	-.061	.002	.601	.230
Czech Republic	24	130	9.53	18.78	.201	.003	.998	.102
Dominican Rep.	33	175	9.92	28.90	-.034	.002	.863	.102
Ecuador	33	218	12.21	12.79	.169	.009	.301	.286
Egypt, Arab Rep.	30	181	8.49	18.43	.517	.002	.828	.135
Guatemala	32	205	12.14	14.46	.808	.008	.803	.614
Honduras	20	122	11.27	8.90	.723	.016	.599	.866
Hong Kong	34	145	10.82	12.20	.228	.002	.754	.962
Hungary	23	130	12.65	22.01	.709	.003	.929	.113
India	77	476	15.36	12.47	.477	.003	.628	.853
Indonesia	62	338	10.99	46.15	.038	.002	.310	.685
Kenya	36	195	8.50	20.65	.194	.003	.897	.947
Korea, Rep.	21	141	10.34	42.61	-.296	.001	.408	.244
Latvia	21	143	15.76	13.78	.720	.021	.359	.264
Lebanon	37	182	7.18	17.01	.790	.038	.978	.137
Malaysia	42	222	12.12	12.72	.122	.006	.944	.553
Mexico	38	222	4.96	18.42	.512	.001	.625	.453
Nigeria	58	309	23.84	22.31	.316	.014	.801	.142
Pakistan	26	158	15.17	25.66	.578	.002	.857	.153

Table 1 (cont'd)

	Banks	Obs.	ROE mean	ROE s.d.	$\hat{\lambda}_j$	se($\hat{\lambda}_j$)	Hansen	AR(2)
Panama	71	321	15.65	16.29	.377	.015	.417	.330
Paraguay	20	120	14.56	18.97	.573	.014	.968	.282
Peru	18	107	8.43	14.49	.328	.048	1.000	.028
Poland	48	202	8.65	14.90	.594	.008	.972	.553
Romania	24	147	4.31	22.51	.290	.014	.733	.484
Russia	695	2341	12.44	11.83	.315	.020	.172	.290
Senegal	32	138	3.94	26.95	.526	.001	.596	.235
Singapore	16	71	11.05	21.23	.243	.044	.803	.447
Taiwan	46	301	-0.80	17.44	.325	.022	.356	.101
Tanzania	15	52	19.99	26.02	.254	.156	.077	.363
Thailand	21	135	-4.29	55.57	.967	.002	.403	.259
Tunisia	15	94	6.95	7.12	.630	.028	.840	.034
Ukraine	33	188	11.71	11.34	.454	.017	.599	.796
Uruguay	44	195	-9.10	82.89	.589	.010	.692	.665
Venezuela	49	268	25.37	18.63	.391	.025	.358	.593
Vietnam	22	128	10.38	14.59	.580	.010	.479	.271

Note:
λ is estimated by applying the system GMM estimator to (4). Hansen is the p-value for the Hansen test for the validity of the over-identifying restrictions. AR(2) is the p-value for the test for second-order autocorrelation.

Table 2 Frequency distributions of estimated short-run persistence coefficients

	Number of Countries		
	Advanced	Developing	Total
$\lambda_j<0$	0	3	3
$0\le\lambda_j<0.2$	2	6	8
$0.2\le\lambda_j<0.4$	7	11	18
$0.4\le\lambda_j<0.6$	5	17	22
$0.6\le\lambda_j$	5	9	14
Total	19	46	65
Mean λ_j	0.442	0.426	0.430

Table 3 Summary statistics for return on average equity, estimated persistence of profit, and covariates

	Advanced countries			Developing countries			All countries		
	N	Mean	St.dev	N	Mean	St.dev	N	Mean	St.dev
ROE	19	7.99	5.44	46	10.96	5.79	65	10.10	5.81
Persistence of profit, λ_j	19	0.47	0.2	46	0.41	0.26	65	0.43	0.24
Inflation	19	0.02	0.01	44	0.11	0.15	63	0.08	0.13
Market capitalization	19	0.95	0.56	42	0.36	0.58	61	0.54	0.63
GDP growth	19	0.03	0.01	44	0.05	0.02	63	0.04	0.02
Total entry denied	18	0.01	0.02	31	0.10	0.16	49	0.07	0.13
Domestic entry denied	15	0.01	0.04	16	0.17	0.25	31	0.09	0.19
Foreign entry denied	16	0.00	0.01	27	0.09	0.16	43	0.06	0.13
Activity restrictions	19	2.21	0.56	37	2.73	0.63	56	2.55	0.65
Capital requirements	18	0.08	0.00	37	0.09	0.02	55	0.09	0.01
Financial freedom	19	34.78	13.85	42	51.17	13.59	61	46.07	15.57
Economic freedom	19	70.63	5.83	42	60.50	8.46	61	63.65	9.02
KKZ index	19	1.46	0.32	45	-0.06	0.67	64	0.39	0.91
GDP per capita	19	10.34	0.34	43	7.86	1.00	62	8.62	1.43
Property rights	19	84.02	8.73	42	48.42	18.63	61	59.51	23.16
HHI	19	0.16	0.09	46	0.20	0.14	65	0.19	0.13
H-statistic	19	0.52	0.20	44	0.58	0.28	63	0.56	0.26

Note:
See Appendix for variable definitions.

Table 4 Correlation matrix for covariates

	Inflation	Market cap.	GDP growth	Total entry denied	Domestic entry denied	Foreign entry denied	Activity restr.	Capital req.	Financial freedom	Economic freedom	KKZ	GDP per capita	Property rights	HHI
Market capitalization	-0.28													
	0.03													
GDP growth	0.08	-0.19												
	0.55	*0.14*												
Total entry denied	-0.02	-0.15	0.25											
	0.90	*0.31*	*0.09*											
Domestic entry denied	0.21	-0.04	0.36	0.90										
	0.27	*0.84*	*0.05*	*0.00*										
Foreign entry denied	-0.05	-0.19	0.30	0.81	0.45									
	0.75	*0.23*	*0.06*	*0.00*	*0.02*									
Activity restrictions	0.09	-0.42	0.13	0.16	0.18	0.12								
	0.51	*0.00*	*0.34*	*0.28*	*0.34*	*0.45*								
Capital requirements	0.54	-0.39	0.18	0.06	0.05	0.15	0.05							
	0.00	*0.00*	*0.19*	*0.71*	*0.77*	*0.36*	*0.69*							
Financial freedom	0.24	-0.51	0.30	0.31	0.51	0.17	0.45	0.32						
	0.07	*0.00*	*0.02*	*0.03*	*0.00*	*0.29*	*0.00*	*0.02*						
Economic freedom	-0.39	0.71	-0.33	-0.34	-0.42	-0.32	-0.35	-0.53	-0.80					
	0.00	*0.00*	*0.01*	*0.02*	*0.02*	*0.04*	*0.01*	*0.00*	*0.00*					
KKZ index	-0.31	0.59	-0.39	-0.37	-0.38	-0.36	-0.49	-0.45	-0.67	0.79				
	0.01	*0.00*	*0.00*	*0.01*	*0.04*	*0.02*	*0.00*	*0.00*	*0.00*	*0.00*				
GDP per capita	-0.28	0.57	-0.48	-0.42	-0.42	-0.51	-0.52	-0.45	-0.67	0.75	0.91			
	0.03	*0.00*	*0.00*	*0.00*	*0.02*	*0.00*	*0.00*	*0.00*	*0.00*	*0.00*	*0.00*			
Property rights	-0.36	0.62	-0.45	-0.29	-0.26	-0.34	-0.43	-0.57	-0.65	0.85	0.92	0.86		
	0.01	*0.00*	*0.00*	*0.05*	*0.16*	*0.03*	*0.00*	*0.00*	*0.00*	*0.00*	*0.00*	*0.00*		
HHI	-0.02	-0.12	0.12	0.10	0.20	-0.08	0.12	0.07	0.12	-0.13	-0.09	-0.15	-0.11	
	0.88	*0.37*	*0.34*	*0.48*	*0.27*	*0.62*	*0.37*	*0.63*	*0.34*	*0.31*	*0.46*	*0.23*	*0.41*	
H-statistic	0.02	-0.05	-0.17	0.04	0.01	-0.04	-0.05	-0.08	0.00	0.12	0.03	0.07	0.08	0.06
	0.91	*0.72*	*0.20*	*0.81*	*0.95*	*0.82*	*0.70*	*0.58*	*1.00*	*0.36*	*0.82*	*0.57*	*0.57*	*0.62*

Note:
p-values are reported in italics. See Appendix for variable definitions.

Table 5 Second-stage estimation results: macroeconomic and regulatory control covariates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Constant	0.682***	0.742***	1.003***	0.752***	0.740**	0.926**	0.665**
	(0.001)	(0.000)	(0.000)	(0.002)	(0.016)	(0.020)	(0.030)
Inflation	0.721	0.830	-4.000*	0.631	0.794	-0.524	0.715
	(0.343)	(0.255)	(0.057)	(0.472)	(0.334)	(0.332)	(0.353)
Market capitalization	-0.017	0.035	-0.071	0.020	-0.029	-0.080*	-0.014
	(0.803)	(0.604)	(0.715)	(0.801)	(0.709)	(0.091)	(0.871)
GDP growth	-11.592**	-17.625***	-12.838***	-16.440***	-11.182*	4.379	-11.682**
	(0.011)	(0.000)	(0.010)	(0.006)	(0.051)	(0.299)	(0.014)
Total entry denied		1.499***					
		(0.000)					
Domestic entry denied			0.867***				
			(0.000)				
Foreign entry denied				1.326**			
				(0.028)			
Activity restrictions					-0.029		
					(0.655)		
Capital requirements						-6.979	
						(0.177)	
Financial freedom							0.000
							(0.942)
N	59	45	30	39	53	51	59
adj. R-sq	0.082	0.320	0.504	0.141	0.036	0.018	0.065

Note:
The dependent variable is $\hat{\lambda}_j$, as reported in Table 1. See Appendix for covariate definitions. The estimation method is Weighted Least Squares, with weights defined as the reciprocal of the estimated variance of $\hat{\lambda}_j$. Robust standard errors are used to compute the t-statistics, for which p-values are reported in parentheses. *,** and *** denote coefficients significantly different from zero (two-tail tests) at the 0.1, 0.05 and 0.01 levels, respectively.

Table 6 Second-stage estimation results: macroeconomic, institutional and external governance, and market structure and competition covariates

	(8)	(9)	(10)	(11)	(12)	(13)
Constant	2.214***	1.227***	1.914***	0.869***	0.445**	1.069***
	(0.004)	(0.000)	(0.000)	(0.000)	(0.040)	(0.000)
Inflation	0.172	-0.045	0.008	0.274	0.903	0.339
	(0.825)	(0.957)	(0.991)	(0.731)	(0.223)	(0.640)
Market capitalization	0.162	0.034	0.026	0.012	-0.001	-0.041
	(0.133)	(0.634)	(0.696)	(0.865)	(0.993)	(0.538)
GDP growth	-16.178***	-16.039***	-18.099***	-15.011***	-9.437**	-13.792***
	(0.001)	(0.001)	(0.000)	(0.003)	(0.034)	(0.002)
Economic freedom	-0.022**					
	(0.034)					
KKZ index		-0.006**				
		(0.043)				
GDP per capita			-0.104			
			(0.115)			
Property rights				-0.108***		
				(0.007)		
HHI					0.596**	
					(0.032)	
H-statistic						-0.333**
						(0.010)
N	59	59	59	59	59	57
adj. R-sq	0.140	0.134	0.184	0.108	0.142	0.208

Note:
See note to Table 5.